ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. ME ENGINEERING â€" SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) Define Thermodynamic System, Surrounding, and Boundary. Differentiate between intensive and extensive properties. [7 Marks]
(b) State Zeroth Law and First Law of Thermodynamics for a closed system undergoing a cycle and a change of state. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) Derive Steady Flow Energy Equation (SFEE) from first principles. Apply SFEE to Nozzle, Turbine, and Compressor. [7 Marks]
(b) 1 kg of air at 1 bar and 300 K is compressed polytropically (n=1.3) to 6 bar. Find work done, heat transfer, and change in internal energy. [7 Marks]

--- UNIT II ---

Q.2 (a) State Kelvin-Planck and Clausius statements of Second Law of Thermodynamics. Prove their equivalence. [7 Marks]
(b) State and prove Carnot's Theorem. Derive efficiency of Carnot heat engine operating between temperatures T1 and T2. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) State Clausius Inequality and prove that entropy of an isolated system always increases (Principle of Increase of Entropy). [7 Marks]
(b) A reversible heat engine absorbs 800 kJ from 800 K source and rejects heat to 300 K sink. Find thermal efficiency and heat rejected. [7 Marks]

--- UNIT III ---

Q.3 (a) Define Availability, Exergy, and Anergy. Derive expression for availability of closed system and open system. [7 Marks]
(b) What is Irreversibility? State and prove Gouy-Stodola theorem. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Derive Maxwell's Thermodynamic Relations from characteristic state equations. [7 Marks]
(b) Derive Clausius-Clapeyron equation for liquid-vapor phase change and Joule-Thomson coefficient equation. [7 Marks]

--- UNIT IV ---

Q.4 (a) Explain formation of steam with temperature-enthalpy and P-v-T diagrams. Define Dryness Fraction. [7 Marks]
(b) Describe Throttling Calorimeter and Combined Separating & Throttling Calorimeter for measuring dryness fraction. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) Explain Rankine Cycle with T-s and h-s (Mollier) diagrams. Derive expression for Rankine cycle efficiency. [7 Marks]
(b) Describe Reheat Cycle and Regenerative Feedwater Heating Cycle with thermal efficiency advantages. [7 Marks]

--- UNIT V ---

Q.5 (a) Derive air-standard efficiency and mean effective pressure (MEP) for Otto cycle with P-v and T-s diagrams. [7 Marks]
(b) An engine working on Diesel cycle has compression ratio 15 and cut-off takes place at 5% of stroke. Find air-standard efficiency. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Compare Otto, Diesel, and Dual combustion cycles on the basis of same compression ratio and same peak pressure. [7 Marks]
(b) Explain Brayton cycle for gas turbines with regeneration, intercooling, and reheating. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****